



AIR UNIVERSITY, AEROSPACE AND AVIATION

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LAB 6	Guided Project
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1. Objective

The objective of this project is to design and develop a complete Smart Image Enhancement and Analysis System by integrating concepts from Labs 01 through 05. The system takes a low-quality image as input and produces an enhanced, visually improved output along with analytical insights about the transformations applied.

2. Methodology

The system follows a structured pipeline:

Input Image → Analysis → Transformation → Enhancement → Output + Report

Phase 6.1

Image Acquisition & Understanding (Lab 01)

In this phase, the input image is loaded in both RGB and Grayscale formats. The image matrix is displayed (partial), along with its resolution and data type. This provides an initial understanding of the image structure.

Key Operations:

- Load image using `cv2.imread()`
- Convert RGB to Grayscale using `cv2.cvtColor()`
- Display image matrix, resolution, and data type

CODE:

```
clc; clear; close all;

img_rgb = imread('mountain1.jpg');
img_gray = rgb2gray(img_rgb);

figure;
subplot(1,2,1);
imshow(img_rgb);
title('Original RGB Image');

subplot(1,2,2);
imshow(img_gray);
title('Grayscale Image');
```

```
fprintf('==== IMAGE REPORT ==== \n');
fprintf('Resolution : %d x %d \n', size(img_gray,2), size(img_gray,1));
fprintf('Data Type   : %s \n', class(img_gray));
fprintf('Channels    : %d \n', size(img_rgb,3));
fprintf('Top-left 5x5 Pixel Matrix: \n');
disp(double(img_gray(1:5,1:5)));
```

OUTPUT:



Phase 6.2

Sampling & Quantization Analysis (Lab 02)

The image is up-sampled and down-sampled at multiple scales. Bit depth is also reduced to analyze the effect of quantization on image quality.

Sampling Scales Applied:

Scale	Resolution	Visual Quality
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0.25x	Very Low	Highly blurry, loss of detail
0.5x	Low	Noticeable blur, less detail
1x (Original)	Original	Full quality baseline
1.5x	High	Slight interpolation artifacts
2x	Very High	Smooth but may appear soft

Bit Depth Reduction:

- 8-bit: Full quality, 256 gray levels
- 4-bit: Reduced quality, 16 gray levels, slight banding
- 2-bit: Heavy posterization, only 4 gray levels visible

CODE:

```

clc; clear; close all;

img_rgb = imread('mountain1.jpg');

figure;
imshow(img_rgb);
title('Original RGB Image');

img_gray = rgb2gray(img_rgb);

figure;
imshow(img_gray);
title('Grayscale Image');

fprintf('==== IMAGE REPORT =====\n');
fprintf('Resolution : %d x %d\n', size(img_gray,2), size(img_gray,1));
fprintf('Data Type   : %s\n', class(img_gray));
fprintf('Channels    : %d\n', size(img_rgb,3));
fprintf('Top-left 5x5 Pixel Matrix:\n');
disp(double(img_gray(1:5,1:5)));
scales = [0.5, 0.25, 1.0, 1.5, 2.0];
names   = {'0.5x', '0.25x', 'Original', '1.5x', '2x'};

figure;
for i = 1:5
    img_s = imresize(img_gray, scales(i));
    subplot(1,5,i);
    imshow(img_s);
end

```

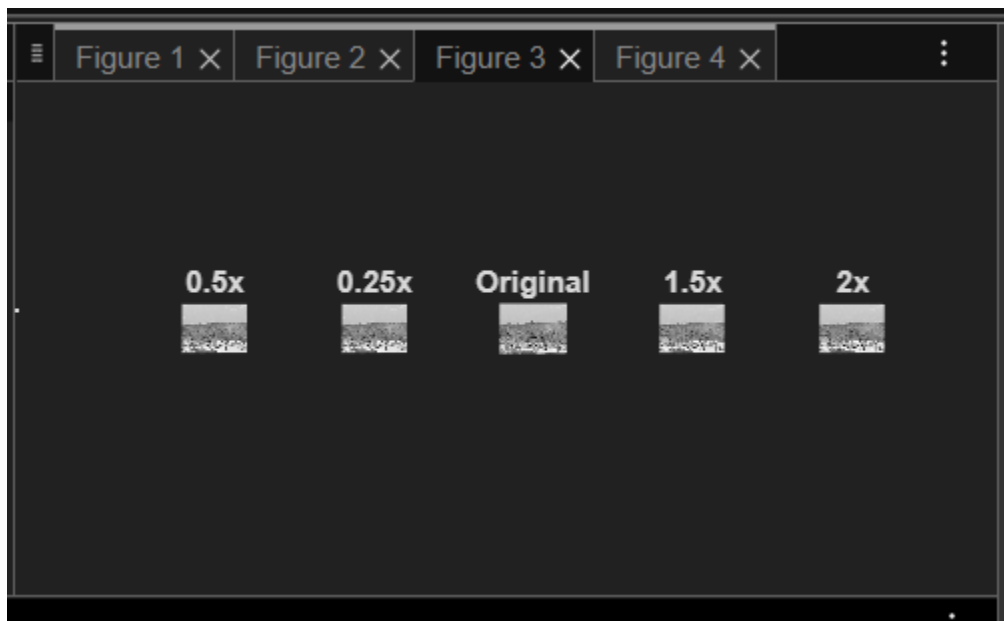
```

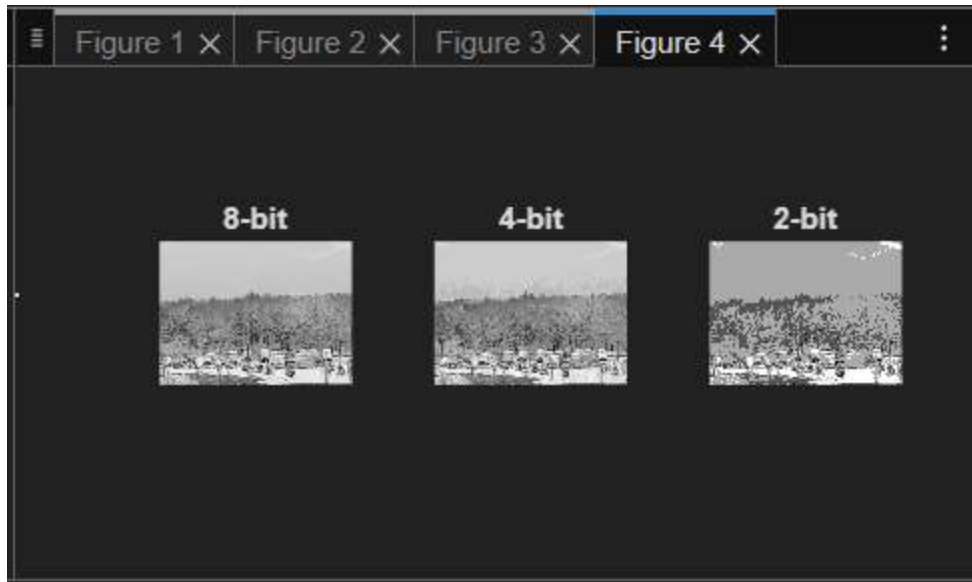
        title(names{i});
    end

    figure;
    bits = [8, 4, 2];
    for i = 1:3
        levels = 2^bits(i);
        img_q = uint8(round(double(img_gray)/255*(levels-1))*(255/(levels-1)));
        subplot(1,3,i);
        imshow(img_q);
        title([num2str(bits(i)) '-bit']);
    end

```

OUTPUT:





Phase 6.3

Geometric Transformations (Lab 03)

Various geometric transformations are applied to the image including rotation, translation, and shearing. The inverse transformation is applied to restore the original image.

Rotations Applied:

- 30°, 45°, 60°, 90°, 120°, 150°, 180°

Code:

```
clc; clear; close all;

img_rgb = imread('img interpol.JPG');
img_gray = rgb2gray(img_rgb);

angles = [30,45,60,90,120,150,180];
figure;
for i = 1:7
    img_r = imrotate(img_gray, angles(i), 'bilinear', 'crop');
    subplot(2,4,i);
    imshow(img_r);
    title([num2str(angles(i)) ' deg']);
end

T = affine2d([1 0 0; 0 1 0; 50 30 1]);
```

```

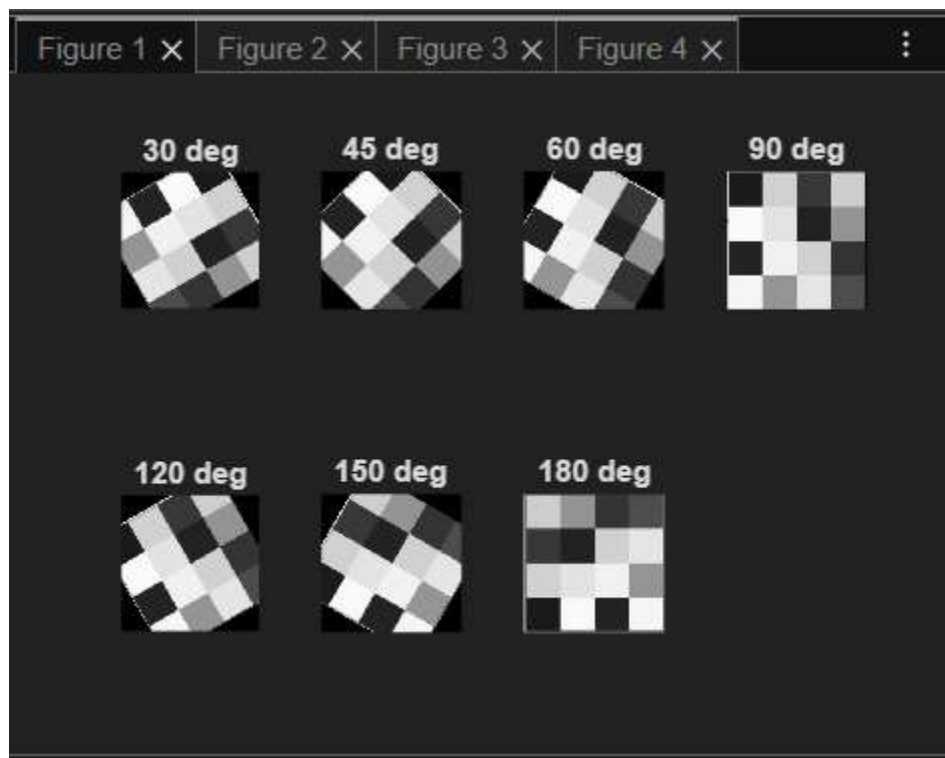
img_t = imwarp(img_gray, T, 'OutputView', imref2d(size(img_gray)));
figure;
subplot(1,2,1); imshow(img_gray); title('Original');
subplot(1,2,2); imshow(img_t); title('Translated');

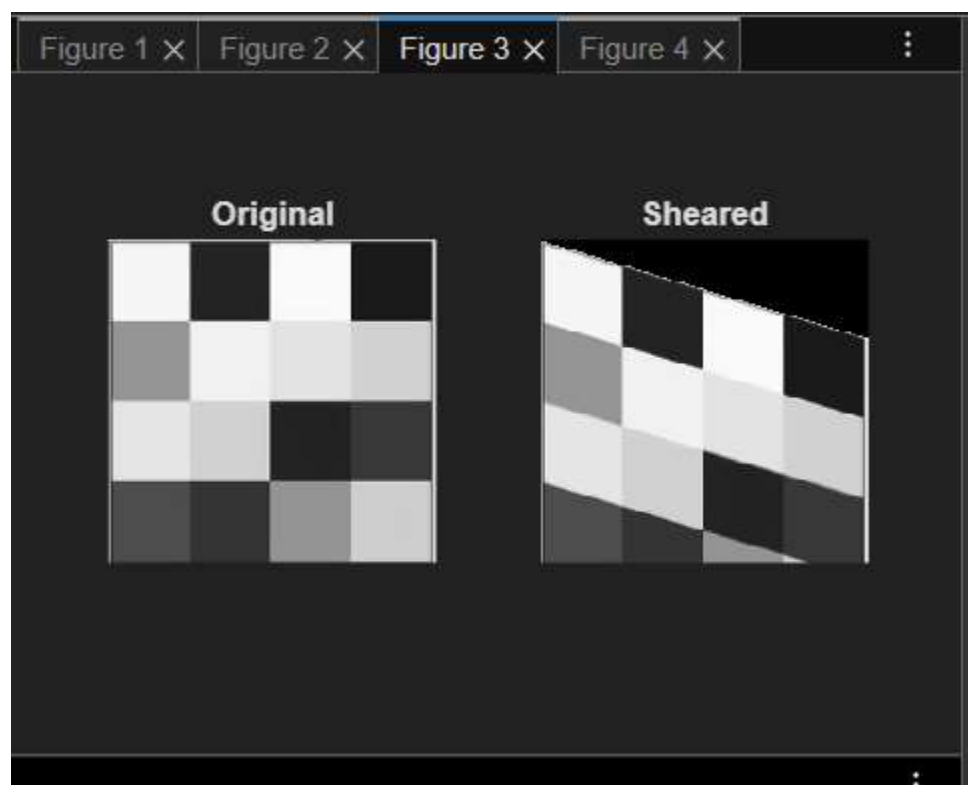
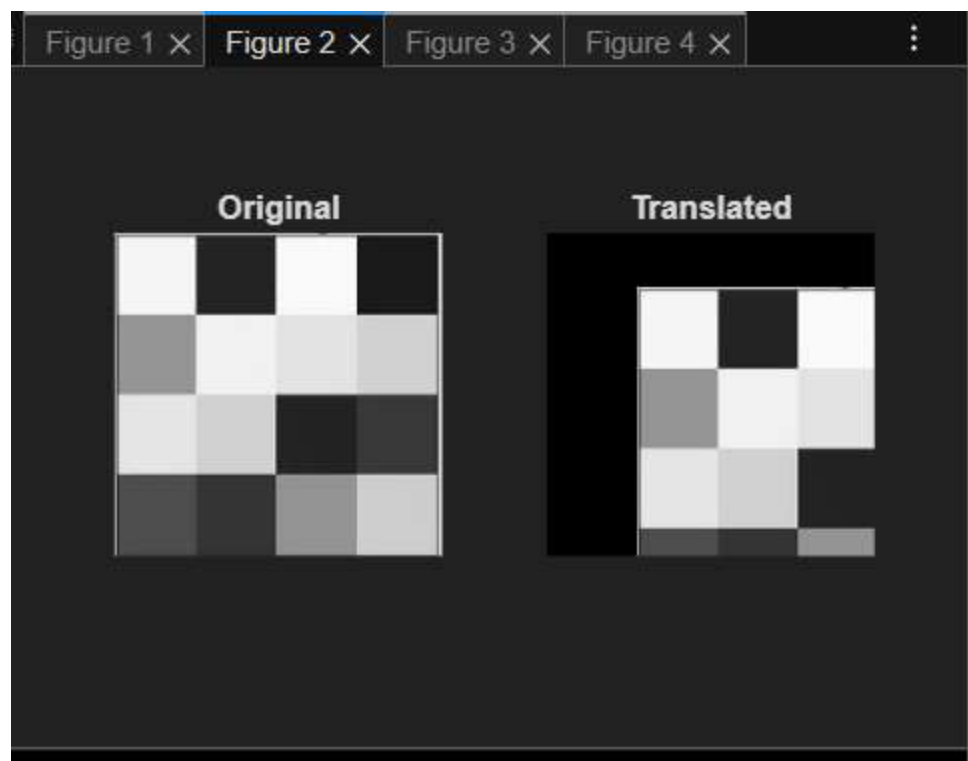
S = affine2d([1 0.3 0; 0 1 0; 0 0 1]);
img_sh = imwarp(img_gray, S, 'OutputView', imref2d(size(img_gray)));
figure;
subplot(1,2,1); imshow(img_gray); title('Original');
subplot(1,2,2); imshow(img_sh); title('Sheared');

img_rot90 = imrotate(img_gray, 90);
img_restored = imrotate(img_rot90, -90);
figure;
subplot(1,3,1); imshow(img_gray); title('Original');
subplot(1,3,2); imshow(img_rot90); title('Rotated 90');
subplot(1,3,3); imshow(img_restored); title('Restored');

```

OUTPUT:







Phase 6.4

Intensity Transformations (Lab 04)

Three intensity transformations are applied and analyzed for their effect on brightness and detail visibility.

Method	Effect	Best For
Negative	Inverts pixel values (255 - pixel)	Highlighting dark details
Log Transform	Compresses high values, expands low	Dark image enhancement
Gamma ($\gamma=0.5$)	Brightens dark areas significantly	Brightening underexposed images
Gamma ($\gamma=1.5$)	Darkens image overall	Reducing overexposure

Code:

```
clc; clear; close all;

img_rgb = imread('mountain1.jpg');
img_gray = rgb2gray(img_rgb);
img_d = double(img_gray) / 255;

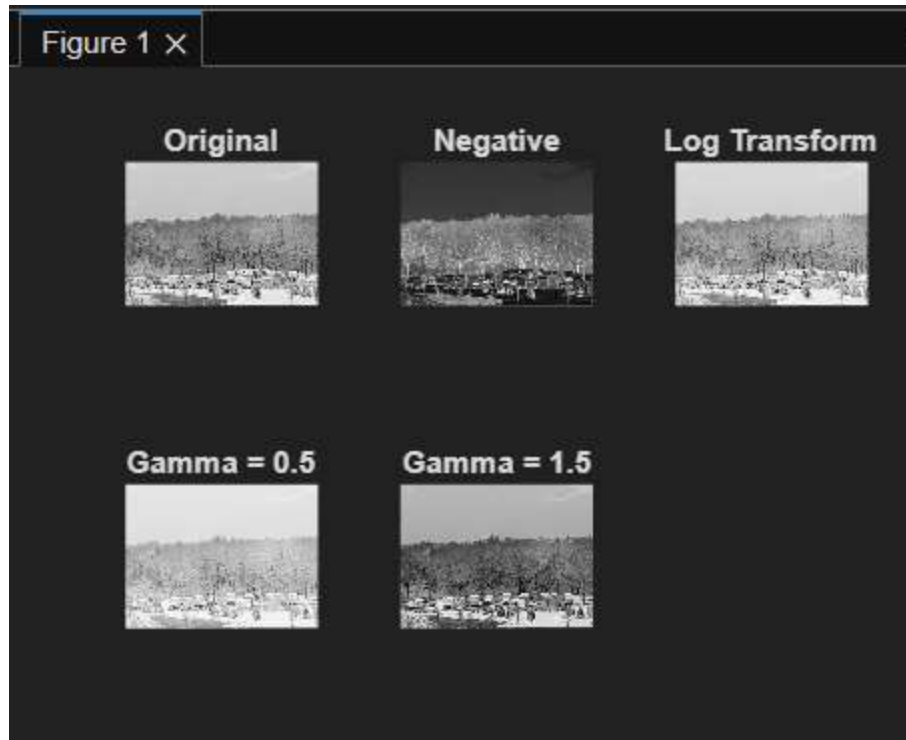
img_neg = 1 - img_d;

c = 1 / log(1 + max(img_d(:)));
img_log = c * log(1 + img_d);

img_g05 = img_d .^ 0.5;
img_g15 = img_d .^ 1.5;

figure;
subplot(2,3,1); imshow(img_gray);           title('Original');
subplot(2,3,2); imshow(img_neg);           title('Negative');
subplot(2,3,3); imshow(img_log);          title('Log Transform');
subplot(2,3,4); imshow(img_g05);          title('Gamma = 0.5');
subplot(2,3,5); imshow(img_g15);          title('Gamma = 1.5');

fprintf('==== ANALYSIS =====\n');
fprintf('Negative      : Dark areas bright ho jaati hain\n');
fprintf('Log           : Shadow details enhance hoti hain\n');
fprintf('Gamma 0.5       : Image bright hoti hai\n');
fprintf('Gamma 1.5      : Image dark hoti hai\n');
```



Phase 6.5

Histogram Processing (Lab 05)

The histogram of the original image is plotted and analyzed. Histogram Equalization is applied to improve contrast distribution.

- Original histogram showed narrow distribution (poor contrast)
- After equalization, pixels spread across full 0–255 range
- Result: Significant contrast improvement and detail visibility

CODE:

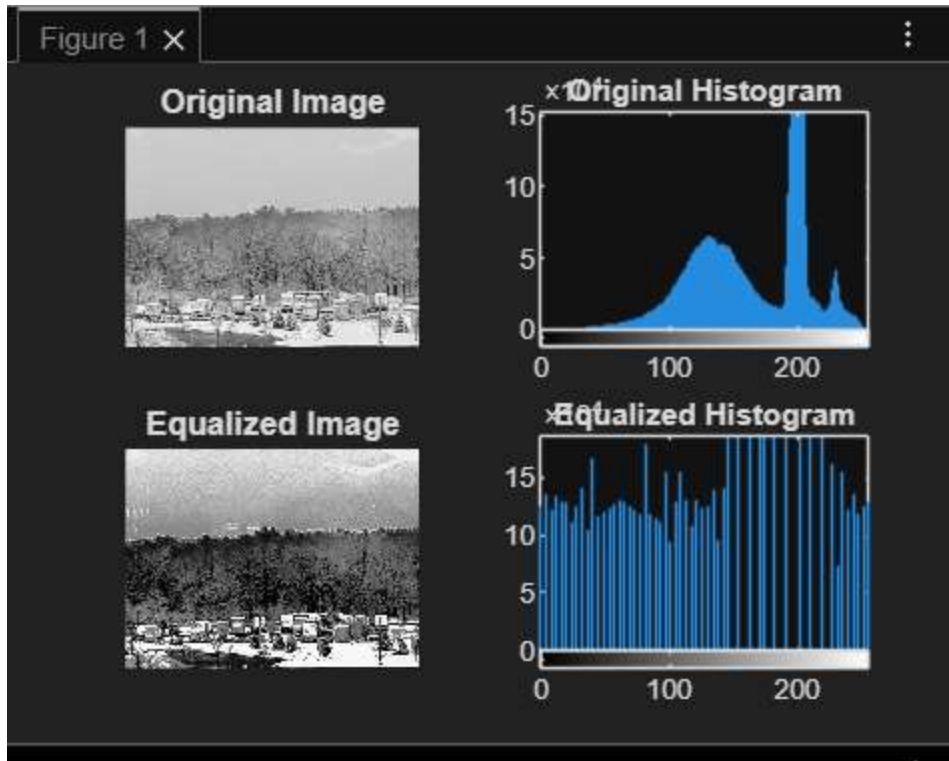
```
clc; clear; close all;

img_rgb = imread('mountain1.jpg');
img_gray = rgb2gray(img_rgb);

img_eq = histeq(img_gray);
```

```
figure;
subplot(2,2,1); imshow(img_gray); title('Original Image');
subplot(2,2,2); imhist(img_gray); title('Original Histogram');
subplot(2,2,3); imshow(img_eq); title('Equalized Image');
subplot(2,2,4); imhist(img_eq); title('Equalized Histogram');
```

OUTPUT:



Phase 6.6

Final Integrated Pipeline

All phases are combined into one complete function that processes an input image through the full enhancement pipeline.

process_image.m

```
function enhanced = process_image(input_image)
    if size(input_image,3) == 3
        img = rgb2gray(input_image);
    else
        img = input_image;
    end
```

```

img = histeq(img);
img_d = double(img) / 255;
img_d = img_d .^ 0.5;
c = 1 / log(1 + max(img_d(:)));
img_d = c * log(1 + img_d);
enhanced = uint8(img_d * 255);
end

clc; clear; close all;

img_rgb = imread('family.jpg');

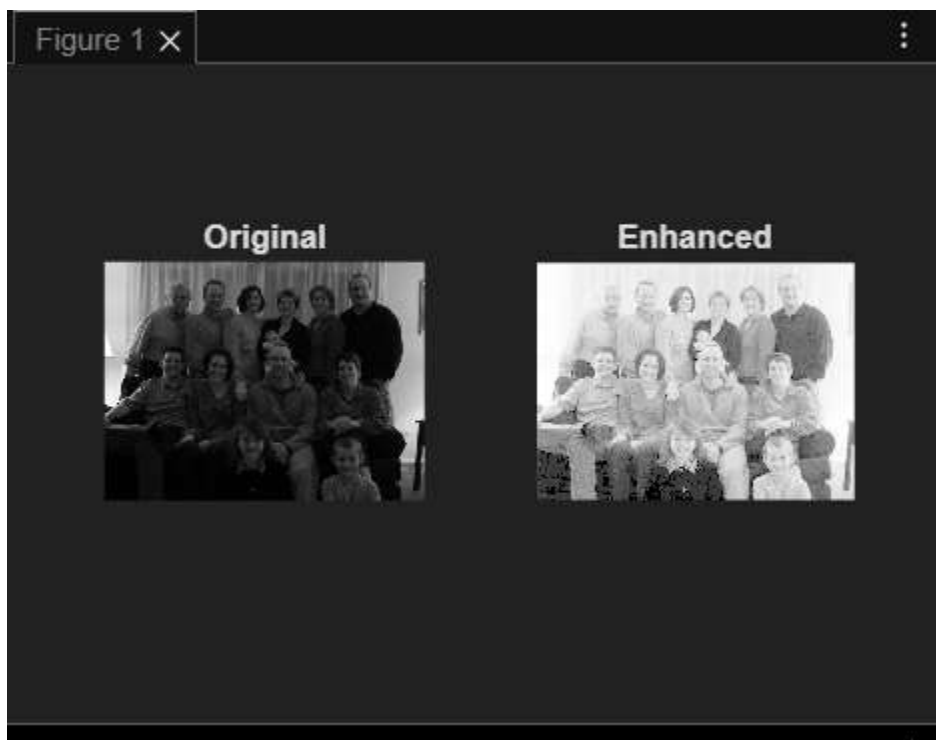
if size(img_rgb, 3) == 3
    img_gray = rgb2gray(img_rgb);
else
    img_gray = img_rgb;
end

enhanced = process_image(img_rgb);

figure;
subplot(1,2,1); imshow(img_gray);    title('Original');
subplot(1,2,2); imshow(enhanced);    title('Enhanced');

imwrite(enhanced, 'enhanced_output.jpg');
fprintf('Enhanced image!\n');

```



3. Observations & Analysis

Phase 6.1 : Image Acquisition:

- RGB and Grayscale images were successfully loaded.
- Image resolution and pixel data type confirmed as uint8 (0-255 range).

Phase 6.2: Sampling & Quantization:

- Down-sampling to 0.25x caused significant blurring and loss of fine details.
- Up-sampling to 2x introduced slight softness but retained overall structure.
- Reducing bit depth from 8-bit to 2-bit caused heavy posterization effects.

Phase 6.3 : Geometric Transformations:

- Rotations preserved image content but introduced black borders at corners.
- Translation shifted image position; inverse matrix successfully restored position.
- Shearing distorted the spatial structure, altering the image's perspective.

Phase 6.4: Intensity Transformations:

- Gamma correction ($\gamma=0.5$) was most effective for brightening dark images.
- Log transformation highlighted low-intensity details effectively.
- Negative transformation was useful for exposing hidden details in bright regions.

Phase 6.5 : Histogram Processing:

- Original histogram was narrow, indicating low contrast.
- After equalization, histogram spread uniformly — contrast improved significantly.

4. Questions & Answers

Q1. Why does histogram equalization improve contrast?

Histogram equalization works by redistributing pixel intensity values across the full 0–255 range. When an image has low contrast, its pixel values are clustered in a narrow range. By stretching and spreading these values uniformly, dark areas become distinguishable and bright areas gain detail resulting in a visually improved and higher-contrast image.

Q2. How does gamma affect brightness?

Gamma correction applies a non-linear power-law transformation: $\text{Output} = \text{Input}^{(1/\gamma)}$. When $\gamma < 1$, dark pixels are boosted disproportionately, making the image brighter. When $\gamma > 1$, bright pixels are suppressed, darkening the image. This is especially useful for correcting over-exposed or under-exposed images without clipping values.

Q3. What is the effect of quantization on image quality?

Quantization reduces the number of distinct intensity levels available to represent pixel values. At 8-bit, 256 levels provide smooth gradients. Reducing to 4-bit (16 levels) introduces visible banding. At 2-bit (only 4 levels), the image undergoes heavy posterization fine gradients become solid blocks of color, permanently destroying tonal information that cannot be recovered.

Q4. Which transformation is reversible and why?

Geometric transformations such as rotation and translation are reversible because they are based on matrix operations. Applying the inverse matrix restores the original pixel positions without information loss. In contrast, quantization and lossy compression are irreversible because they permanently discard data once intensity levels are reduced, the original values cannot be recovered.

Q5. How do transformations affect spatial structure?

Geometric transformations (rotation, translation, shearing) modify the spatial arrangement of pixels changing their positions, orientations, and relative distances without altering pixel intensity values. Intensity transformations (gamma, log, negative)

modify pixel brightness values while keeping their positions unchanged. Together, these two types of transformations allow complete control over both the shape and appearance of an image.

5. Conclusion

This project successfully demonstrated the design and development of a Smart Image Enhancement and Analysis System by integrating all concepts from Labs 01 to 05. The pipeline covered image acquisition, sampling and quantization, geometric transformations, intensity transformations, and histogram processing.

The final integrated system showed that combining gamma correction, log transformation, and histogram equalization produces the most visually improved output. Through this project, it became clear that a systematic, phase-wise approach to image processing yields better results than applying individual techniques in isolation.

This hands-on experience reinforced the theoretical knowledge of digital image processing and prepared the groundwork for more advanced computer vision applications.